Week 5 Cisco

Screenshots

Task 28:

```
Router(config)#hostname R1
R1(config)#
```

Task 29:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#
```

Task 30:

```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z
Switch(config)#hostname SW1
SW1(config)#
```

Task 36:

```
R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
SW1              Gig 0/0           165           S       2960      Fas 0/1
R1#
SW1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
R1               Fas 0/1           136           R       C2900     Gig 0/0
R2               Fas 0/2           155           R       C2900     Gig 0/0
SW1#
R2#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
SW1              Gig 0/0           141           S       2960      Fas 0/2
R2#
```

Task 37:

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z
R1(config)#interface g0/0
R1(config-if)#no cdp enable
R1(config-if)#exit
R1(config)#
```

Task 38:

```
R1(config)#no cdp run
R1(config)#cdp run
R1(config)#end
R1#
```

Task 39:

```
R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce   Holdtme    Capability   Platform    Port ID
R1#
```

Task 40:

```
SW1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual down                  down
```

Task 41:

```
SW1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual administratively down down
```

Task 42:

```
SW1#show interfaces fa0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0006.2a2d.8a02 (bia 0006.2a2d.8a02
  Description: Link to R2
 BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s
```

Task 43:

```
SW1#show interfaces fa0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0006.2a2d.8a02 (bia 0006.2a2d.8a02)
  Description: Link to R2
 BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 100Mb/s
```

Task 44:

```
SW1#show interfaces fa0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0006.2a2d.8a02 (bia 0006.2a2d.8a02)
  Description: Link to R2
 BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s
```

Task 45:

```
SW1#show interfaces fa0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0006.2a2d.8a02 (bia 0006.2a2d.8a02
  Description: Link to R2
 BW 100000 Kbit, DLY 1000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 10Mb/s
```

Task 46:

```
R2>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.10.10.2      YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

Reflection:

When SW1 was configured for half-duplex while R2 remained at full duplex, the interface stayed up but communication could become unreliable. Symptoms may include slow network performance, collisions, late collisions, frame errors, and packet loss. This happens because a half-duplex device can only send or receive data at one time and uses collision detection, while a full-duplex device can transmit and receive simultaneously without expecting collisions. The two devices therefore operate with different assumptions, leading to transmission errors and reduced performance.